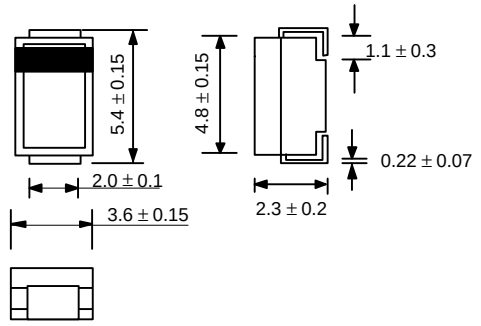
SYNSEMI SEMICONDUCTOR

SK22 - SK2B

SURFACE MOUNT SCHOTTKY BARRIER RECTIFIERS

PRV : 20 - 100 Volts
I_O : 2.0 Amperes

SMB (DO-214AA)



Dimensions in millimeter

FEATURES :

* High current capability
* High surge current capability
* High reliability
* High efficiency
* Low power loss
* Low cost
* Low forward voltage drop

MECHANICAL DATA :

* Case : SMB Molded plastic
* Epoxy : UL94V-O rate flame retardant
* Lead : Lead Formed for Surface Mount
* Polarity : Color band denotes cathode end
* Mounting position : Any
* Weight : 0.1079 gram

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

Rating at 25 °C ambient temperature unless otherwise specified.
Single phase, half wave, 60 Hz, resistive or inductive load.
For capacitive load, derate current by 20%.

RATING	SYMBOL	SK 22	SK 23	SK 24	SK 25	SK 26	SK 27	SK 28	SK 29	SK 2B	UNIT
Maximum Recurrent Peak Reverse Voltage	V_{RRM}	20	30	40	50	60	70	80	90	100	V
Maximum RMS Voltage	V_{RMS}	14	21	28	35	42	49	56	63	70	V
Maximum DC Blocking Voltage	V_{DC}	20	30	40	50	60	70	80	90	100	V
Maximum Average Forward Current See Fig.1	$I_{F(AV)}$	2.0									A
Maximum Peak Forward Surge Current, 8.3ms single half sine wave superimposed on rated load (JEDEC Method)	I_{FSM}	60									A
Maximum Forward Voltage at I_F = 2.0 A. (Note 1)	V_F	0.5			0.74			0.79			V
Maximum Reverse Current at Rated DC Blocking Voltage (Note 1)	I_R	0.5									mA
Junction Temperature Range	T_J	- 65 to + 125			- 65 to + 150						°C
Storage Temperature Range	T_{STG}	- 65 to + 150									°C

Notes :

(1) Pulse Test : Pulse Width = 300 μs, Duty Cycle = 2%.

SYNSEMI SEMICONDUCTOR

RATING AND CHARACTERISTIC CURVES (SK22 - SK2B)

FIG.1 - FORWARD CURRENT DERATING CURVE

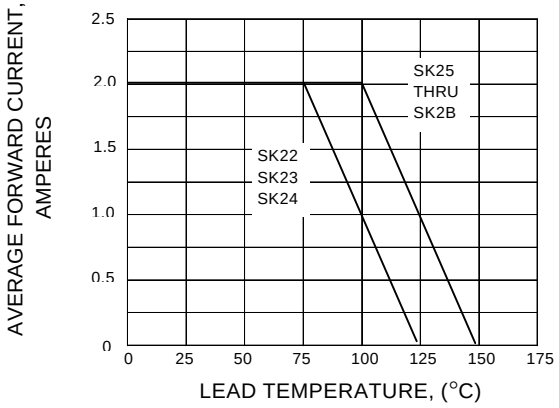


FIG.2 - MAXIMUM NON-REPETITIVE PEAK FORWARD SURGE CURRENT

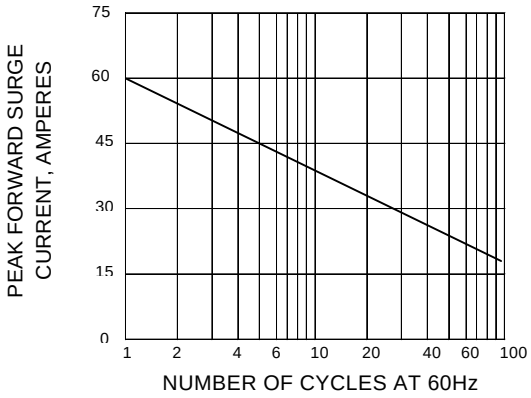


FIG.3 - TYPICAL FORWARD CHARACTERISTICS

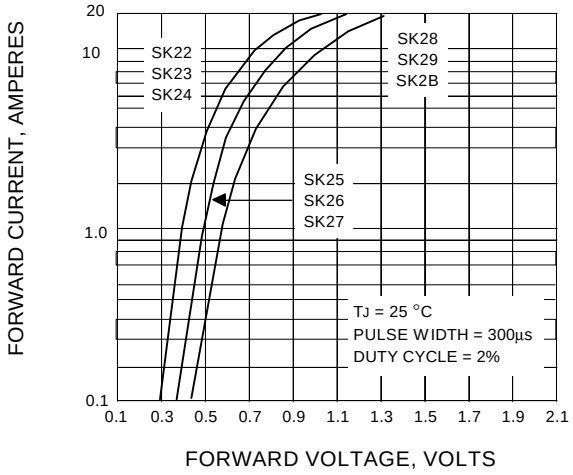
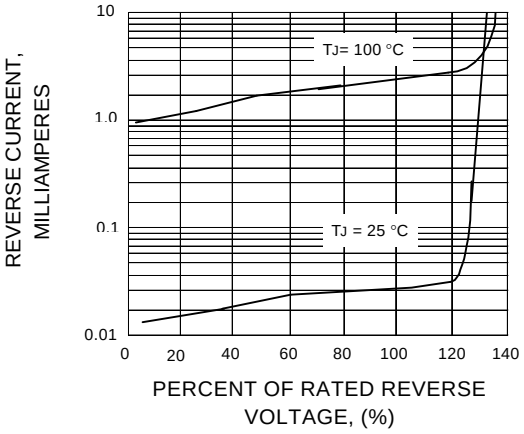



FIG.4 - TYPICAL REVERSE CHARACTERISTICS